FINANCE I

Facultad de Economía y Negocios
Universidad de Chile

Prof.: Alejandro Bernales

Exam

Time Allowed: 1.5 Hours

Instructions:

The solemne has 3 questions, each of them have the same weight.

PLEASE WRITE YOUR NAME IN ALL THE PAGES.

PLEASE ONLY USE THE SPACE IN EACH QUESTION TO ANSWER ALL SECTIONS (USE ONLY ONE SIDE OF THE PAGE - ONE LANGUAGE)

The space provided is sufficient to give a concise and pertinent answer to each question and/or sub-point of a question.

Extra answer booklets will not be provided – please use the spaces provided only. Answers written outside the spaces or on other booklets will not be marked.

Electronic calculators may be used provided that they cannot store text

FINANCE I

Facultad de Economía y Negocios
Universidad de Chile

Prof.: Alejandro Bernales

Examen

Time Allowed: 1.5 Hours

Instrucciones:

El Control tiene tres preguntas, cada una de ellas tienen la misma ponderación.

POR FAVOR ESCRIBA SU NOMBRE EN TODAS LAS HOJAS.

POR FAVOR UTILICE EL ESPACIO PROVISTO EN CADA PREGUNTA PARA RESPONDER CADA SECCIÓN (USAR UN SOLO LADO DE LA HOJA - UN SOLO LENGUAJE).

El espacio entregado es suficiente para entregar una respuesta concisa y pertinente a cada pregunta y/o sub-pregunta.

No se entregarán hojas adicionales para responder por favor utilice solamente el espacio entregado. Respuestas escritas fuera de este espacio o en otras hojas no serán revisadas.

Calculadoras electrónicas pueden ser utilizadas salvo las que puedan almacenar texto

Question I:

Assume that you are a senior financial analyst at an investment bank. One day you get a call from a customer. She wants to know the price of a six-month put option on the stock of Soquimik (Soquimik doesn't paid dividends). The option has a strike price of $ 180 and is European style. The Soquimik's current share price is $182, the risk-free interest rate is 6% per year, and his customer knows that the volatility of the stock is equals to 50%.

1a) What is the price of this put option?
(Hint: Maybe the table at the end of the exam can be useful)

Answer:

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r_f + {\sigma^2}/{2}\right)T}{\sigma\sqrt{T}} = 0{,}2929$$

$$d_{2=}d_1 - \sigma\sqrt{T} = -0{,}06107$$

$$N(d1) = 0{,}6152 \quad N(d2) = 0{,}4758$$

$$N(-d1) = 0{,}3848 \quad N(-d2) = 0{,}5242$$

Remember that $N(-d1) = 1 - N(d1)$

So, the value of the Put options is:

$$Pt = Ke^{-rt} * N(-d2) - St * N(-d1) = 21{,}53$$

Pregunta I:

Asuma que usted es un analista financiero senior en un Banco de inversiones. Un día recibe una consulta de un cliente. Ella quiere saber el precio de una opción de venta a seis meses sobre una acción de Soquimik (Spquimich no paga dividendos). Estas opciones tienen un precio strike de $180 y es del tipo Europea. El precio actual de la acción de Soquimik es de $182, la tasa de interés libre de riesgo es de 6% anual y además sabe que la volatilidad de la acción es igual a 50%.

1a) ¿Cuál es el precio de esta opción de venta?
(Hint: Tal vez la tabla al final del examen puede ser útil)

Respuesta:

1b) What is the price of the call option with same characteristics?
Answer:

So, the value of the Call options is:

$$Ct = St * N(d1) - Ke^{-rt} * N(d2) = 28{,}85$$

1b) ¿cuál es el precio de la opción de compra con las mismas características?

Respuesta

1c) Suppose that the stock will pay a dividend of $ 20 in 5 months. What are the new values for the call and put options?

Answer:

Now change the price of the stock, so we have to calculate the new price of the stock.

$$S_t^* = St - Div * e^{-rt} = 182 - 20 * e^{-0{,}06*\frac{5}{12}} = 162{,}49$$

Now we must re-calculate the new d1 and d2, because St has chances.

$$d_1 = \frac{\ln\left(\frac{S_t^*}{K}\right) + \left(r_f + {\sigma^2}/{2}\right)t}{\sigma\sqrt{t}} = -0{,}0278$$

$$d_{2=}d_1 - \sigma\sqrt{t} = -0{,}3813$$

$$N(d1) = 0{,}4889 \quad N(d2) = 0{,}3515$$

So, the value of the Put options is:

$$Pt = Ke^{-rt} * N(-d2) - S_t^* * N(-d1) = 30{,}24$$

and

$$Ct = S_t^* * N(d1) - Ke^{-rt} * N(d2) = 18{,}05$$

1c) Suponga que la acción pagará un dividendo de $ 20 en 5 meses. ¿Cuáles son los nuevos valores para las opciones de compra y venta ?

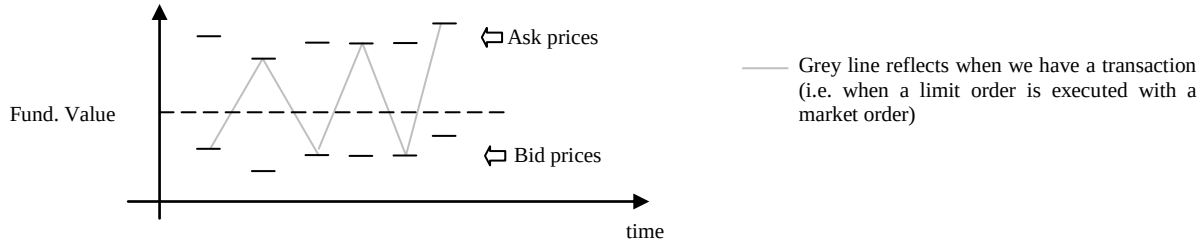
Respuesta

Question II:

2a) When the fundamental value of an asset is constant, then the return volatility of trading prices of the asset is zero. Provide a comment about sentence.

Answer

As explained in classes, this is not true, because agents trade at the bid price and ask prices which are not necessarily equal to the fundamental value of the asset. Bid and ask prices are given by the number of agents who want to wait by submitting limit orders (i.e. they are liquidity providers). They would make profits by the liquidity cost paid by agents who submit market orders, who do not want to wait (i.e. liquidity demanders). Thus, bid and ask prices are not only affected by the fundamental value of the asset, but also by the arrival rate of liquidity providers and liquidity demanders.



2b) In the presentation of the professor "contagio del contagio", he used in a model the credit default swaps (CDS) spreads. How do CDS spreads are related to the interest rates of countries?

Answer:

The CDS spreads measure the default risk of countries, as the interest rates coming from bonds issue by governments.

Pregunta II:

2a) Cuando el valor fundamental de un activo es constante, entonces la volatilidad de los retornos de los precios transados del activo es cero. Comente sobre esta afirmación.

Answer

2b) En la presentación del profesor "contagio del contagio", él utilizó en un modelo los spreads de credit default swaps (CDS). ¿Cómo se relacionan los spreads de CDS con las tasas de interés de los países?

Respuesta:

Question III:

Explain or demonstrate why a Swap can also be valued using FRAs.

Answer:

A FRA is an agreement that a certain specified interest rate, R_k, will apply to a certain principal, L, for a certain specified future time period.

Suppose that the rate observed in the market for the future time period at the beginning of the time period proves to be R_m.

If the FRA is an agreement that R_k will apply when the principal is invested, the holder of the FRA can borrow the principal at R_m and then invest it at R_k.

- The net cash flow at the end of the period is then an inflow of R_k *L and an outflow of R_m *L.

If the FRA is an agreement that R_k will apply when the principal is borrowed, the holder of the FRA can invest the borrowed principal at R_m.

- The net cash flow at the end of the period is then an inflow of R_m *L and an outflow of R_k *L.

In either case we see that the PRA involves the exchange of a fixed rate of interest, R_k, on the principal of L for the floating rate of interest observed in the market, R_m.

Pregunta III:

Explique o demuestre por qué un Swap también puede ser valorizado mediante el uso de FRAs.

Table for $N(x)$ When $x \leqslant 0$

This table shows values of $N(x)$ for $x \leqslant 0$. The table should be used with interpolation. For example,

$$\begin{aligned} N(-0.1234) &= N(-0.12) - 0.34[N(-0.12) - N(-0.13)] \\ &= 0.4522 - 0.34 \times (0.4522 - 0.4483) \\ &= 0.4509 \end{aligned}$$

x	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−3.0	0.0014	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
−3.6	0.0002	0.0002	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
−3.7	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
−3.8	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
−3.9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
−4.0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table for $N(x)$ When $x \geqslant 0$

This table shows values of $N(x)$ for $x \geqslant 0$. The table should be used with interpolation. For example,

$$\begin{aligned} N(0.6278) &= N(0.62) + 0.78[N(0.63) - N(0.62)] \\ &= 0.7324 + 0.78 \times (0.7357 - 0.7324) \\ &= 0.7350 \end{aligned}$$

x	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9986	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
4.0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000